TALGESIL INJECTION 0.1 MG/2 ML

DESCRIPTION: TALGESIL INJECTION 0.1 MG/2 ML is a clear colourless solution.

COMPOSITION: Each ml contains Fentanyl (as Fentanyl Citrate) 0.05 mg

PHARMACODYNAMICS:

Opioid analgesics bind with stereospecific receptors at many sites within the central nervous system (CNS) to alter processes affecting both the perception of and emotional response to pain. Although the precise sites and mechanisms of action have not been fully determined, alterations in the release of various neurotransmitters from afferent nerves sensitive to painful stimuli may be partially responsible for the analgesic effects.

It has been proposed that there are multiple subtypes of opioid receptors, each mediating various therapeutic and/or side effects of opioid drugs. The actions of an opioid analgesic may therefore depend upon whether it acts as a full agonist or a partial agonist or is inactive at each type of receptor. Fentanyl and its derivatives probably produce their effects via agonist actions at the mureceptor.

PHARMACOKINETICS:

Fentanyl citrate is absorbed from the gastro-intestinal tract. After parenteral administration it has a rapid onset and short duration of action. It is rapidly metabolised, probably by oxidative N-dealkylation, and excreted mainly in the urine.

The short duration of action is probably due to redistribution rather than metabolism and excretion. Up to 70% has been reported to be bound to plasma proteins.

Following intravenous injection of 100 µg the effect begins almost immediately, although maximum analgesia and respiratory depression may not occur for several minutes, and the average duration of action is 30 to 60 minutes although analgesia may only last for 10 to 20 minutes.

Increasing the dose to 50 µg per kg body-weight can give pain relief for 4 to 6 hours.

INDICATION: Short duration analgesia during pre-medication induction and maintenance of anaesthesia, and in the immediate post-operative period.

RECOMMENDED DOSAGE:

Dosage should be individualised according to age, bodyweight, physical status, underlying pathological condition, use of other drugs, type of anaesthesia to be used and the surgical procedure involved.

Adults: Premedication: (To be appropriately modified in the elderly, debilitated and those who receive other depressant drugs):

50 to 100 mcg (1 to 2 mL) may be administered intramuscularly 30 to 60 minutes prior to surgery.

Adjunct to general anaesthesia: Induction 50 to 100 mcg (1 to 2 mL) i.v. initially, repeat at two to three minutes intervals until desired effect is achieved. A reduced dose of 25 to 50 mcg (0.5 to 1 mL) is recommended in elderly and poor risk patients.

Maintenance: 25 to 50 mcg (0.5 to 1 mL) i.v. or i.m. when movement and/or changes in vital signs indicate surgical stress or lightening of analgesia.

Adjunct to regional anaesthesia: 50 to 100 mcg (1 to 2 mL) may be administered i.m. or slowly i.v. when additional analgesia is required.

Post-operatively (recovery room): 50 to 100 mcg (1 to 2 mL) may be administered i.m. for the control of pain, tachypnoea and emergence delirium.

The dose may be repeated in one or two hours as needed.

Children: For induction and maintenance in children 2 to 12 years of age, a reduced dose of 20 to 30 mcg (0.4 to 0.6 mL) per 10 kg is recommended.

Impaired renal function: Fentanyl should be used with caution.

Impaired hepatic function: Fentanyl should be used with caution.

ROUTE OF ADMINISTRATION: For i.m. & i.v.injection

CONTRACINDICATIONS:

Known intolerance to fentanyl.

Bronchial asthma.

Head injuries and increased intracranial pressure: As for any narcotic analgesic, fentanyl should not be used in patients susceptible to respiratory depression, such as comatose patients who may have head injuries or a brain tumour (see also Precautions).

Fentanyl may obscure the clinical course of patients with head injury.

Concomitant MAO inhibitors: Severe and unpredictable potentiation by MAO inhibitors has been reported with narcotic analgesics and the use of fentanyl in patients who have received MAO inhibitors within 14 days is not recommended.

Myasthenia gravis: Fentanyl may cause rigidity upon i.v. administration. Therefore, the need for reversal with muscle relaxants contra-indicates its use in patients with a history of myasthenia gravis.

WARNINGS & PRECAUTIONS:

Impaired respiration: Fentanyl should be used with caution in patients with severe impairment of pulmonary function because of the possibility of respiratory depression (eg. chronic obstructive pulmonary disease, patients with decreased respiratory reserve, or any patient with potentially compromised respiration).

In such patients, narcotics may further decrease respiratory drive and increase airway resistance. During anaesthesia, this can be managed by assisted or controlled respiration.

Use of narcotic antagonists for respiratory depression: Respiratory depression caused by narcotic analgesics can be reversed by narcotic antagonists.

However, appropriate surveillance should be maintained because the duration of respiratory depression of doses of fentanyl employed during anaesthesia is usually longer than the duration of narcotic antagonist action. Consult individual production information (nalorphine or naloxone) before employing narcotic antagonists.

Impaired liver kidney function: Fentanyl should be administered with caution to patients with liver and kidney dysfunction because of the importance of these organs in the metabolism and excretion of drugs.

Bradycardia: Fentanyl may produce bradycardia, which may be treated with atropine; however, it should be used with caution in patients with cardiac bradyarrhythmias.

Sphincter of Oddi Spasm: As has been observed with all narcotic analgesics, episodes suggestive of sphincter of Oddi Spasm may occur with Fentanyl.

Adjunct to conduction anaesthesia: Nitrous oxide: Nitrous oxide has been reported to produce cardiovascular depression when given with high doses of fentanyl.

Certain forms of conduction anaesthesia, such as spinal anaesthesia and some peridural anaesthetics, can alter respiration by blocking intercostal nerves. Through other mechanisms fentanyl can also alter respiration.

Therefore, when fentanyl is used to supplement these forms of anaesthesia, the anaesthetist should be familiar with the physiological alterations involved and be prepared to manage them in patients selected for these forms of anaesthesia. When used with a neuroleptic such as droperidol, blood pressure may be altered and hypotension can occur.

Monitoring: Vital signs should be monitored routinely.

Elderly, debilitated patients: The initial fentanyl dose should be reduced.

Warnings: Adequate facilities should be available for post-operative monitoring and ventilation. Resuscitative equipment, oxygen and a narcotic antagonist should be readily available to manage apnoea.

Concomitant neuroleptics: If fentanyl is administered with neuroleptics, the user should be familiar with the special properties of each drug, particularly with regard to duration of action. In addition, when such a combination is used fluids and other counter measures to manage hypotension should be available.

Total narcotic dose: As with other potent narcotics, the respiratory depressant effect of fentanyl persists longer than the measured analgesic effect. The total dose of all narcotic analgesics are given during recovery from anaesthesia. It is recommended that post-operative narcotics, when required, should be used initially in reduced doses, as low as 1/4 to 1/3 of those usually recommended.

Muscle rigidity: Fentanyl may cause rigidity, particularly involving the muscles of respiration. This effect is related to the dose and speed of injection and may be reduced by slow intravenous injection. If this effect occurs, it may be managed by the use of assisted or controlled respiration and, if necessary, by administration of a neuromuscular blocking agent compatible with the patient's conditions.

Drug Dependence: Fentanyl can produce drug dependence of the morphine type and therefore has the potential for being abused.

Respiratory depression: Depression of respiration is the most marked and dangerous side effect of fentanyl. In the post-operative period, patients may exhibit delayed depression of respiration. Patients should be monitored for this possibility and appropriate countermeasures taken as necessary (See **Precautions**).

Risks from Concomitant Use with Benzodiazepines

Profound sedation, respiratory depression, coma, and death may result from the concomitant use of Talgesil Injection with benzodiazepines. Observational studies have demonstrated that concomitant use of opioids and benzodiazepines increases the risk of drug-related mortality compared to use of opioids alone. Because of these risks, reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate.

If the decision is made to newly prescribe a benzodiazepine and an opioid together, prescribe the lowest effective dosages and minimum durations of concomitant use.

If the decision is made to prescribe a benzodiazepine in a patient already receiving an opioid, prescribe a lower initial dose of the benzodiazepine than indicated in the absence of an opioid, and titrate based on clinical response.

Follow patients closely for signs and symptoms of respiratory depression and sedation. Advise both patients and caregivers about the risks of respiratory depression and sedation when Talgesil Injection is used with benzodiazepines. Advise patients not to drive or operate heavy machinery until the effects of concomitant use of the benzodiazepines have been determined. Screen patients for risk of substance use disorders, including opioid abuse and misuse, and warn them of the risk for overdose and death associated with the use of benzodiazepines (See Drug Interactions).

Serotonin Syndrome with Concomitant Use of Serotonergic Drugs

Cases of serotonin syndrome, a potentially life-threatening condition, have been reported during concurrent use of Talgesil Injection with serotonergic drugs (See Interactions with Other Medicaments). This may occur within the recommended dosage range.

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Serotonin syndrome symptoms may include mental-status changes (e.g. agitation, hallucinations, coma), autonomic instability (e.g. tachycardia, labile blood pressure, hyperthermia), neuromuscular aberrations (e.g. hyperreflexia, incoordination) and/or gastrointestinal symptoms (e.g. nausea, vomiting, diarrhoea) and can be fatal (See Interactions with Other Medicaments). The onset of symptoms generally occurs within several hours to a few days of concomitant use, but may occur later that that. Discontinue Talgesil Injection if serotonin syndrome is suspected.

Adrenal Insufficiency

Cases of adrenal insufficiency have been reported with opioid use, more often following greater than one month of use. Presentation of adrenal insufficiency may include non-specific symptoms and signs including nausea, vomiting, decreased appetite, fatigue, weakness, dizziness, and low blood pressure. If adrenal insufficiency is suspected, confirm the diagnosis with diagnostic testing as soon as possible. If adrenal insufficiency is diagnosed, treat with physiological replacement dosing of corticosteroids. Wean the patient off of the opioids to allow adrenal function to recover and continue corticosteroid treatment until adrenal function recovers. Other opioids may be tried as some cases reported use of a different opioid without recurrence of adrenal insufficiency. The information available does not identify any particular opioids as being more likely to be associated with adrenal insufficiency.

Sexual Function/Reproduction

Long term use of opioids may be associated with decreased sex hormone levels and symptoms such as low libido, erectile dysfunction, or infertility (See Postmarketing Experience)

INTERACTION WITH OTHER MEDICAMENTS:

Narcotic supplement to general and regional anaesthesia. Combination with a neuroleptic as an anaesthetic pre-medication for the induction of anaesthesia, and as an adjunct in the maintenance of general and regional anaesthesia.

Other CNS depressants: Other CNS depressants, eg. barbiturates neuroleptics, narcotics and general anaesthetics, will have additive or potentiating effects with fentanyl. When patients have received such drugs, the dose of fentanyl required will be less than usual. Likewise, following the administration of fentanyl the dose of other CNS depressant drugs should be reduced.

Fentanyl is incompatible with thiopentone and methohexitone.

Benzodiazepines

Due to additive pharmacologic effect, the concomitant use of opioids with benzodiazepines increases the risk of respiratory depression, profound sedation, coma and death.

The concomitant use of opioids and benzodiazepines increases the risk of respiratory depression because of actions at different receptor sites in the central nervous system that control respiration. Opioids interact primarily at μ-receptors, and benzodiazepines interact at $GABA_A$ sites. When opioids and benzodiazepines are combined, the potential for benzodiazepines to significantly worsen opioid-related respiratory depression exists.

Reserve concomitant of these drugs for use in patients for whom alternative treatment options are inadequate (see Warnings and Precaustions)

Limit dosage and duration of concomitant use of benzodiazepines and opioids, and follow patients closely for respiratory depression and sedation.

Serotonergic Drugs

The concomitant use of opioids with other drugs that affect the serotonergic neurotransmitter system has resulted in serotonin syndrome. If concomitant use is warranted, carefully observe the patient, particularly during treatment initiation and dose adjustment. Discontinue Talgesil Injection if serotonin syndrome is suspected. Examples of serotonergic drugs are selective serotonin reuptake inhibitors (SSRIs), serotonin and norepinephrine reuptake inhibitors (SNRIs), tricyclic antidepressants (TCAs), triptans, 5-HT3 receptor antagonists, drugs that affect the serotonin neurotransmitter system (e.g. mirtazapine, trazodone, tramadol), momoamine oxidase (MAO) inhibitors (those intended to treat psychiatric disorders and also others, such as linezolid and intravenous methylene blue) (See Warnings and Precautions).

USE IN PREGNANCY & LACTATION:

Use in Pregnancy: Narcotic analgesics may cause respiratory depression in the newborn infant. These products should only be used during labour after weighing the needs of the mother against the risk to the foetus.

SIDE EFFECTS:

Adverse Reactions: **More common**: Respiratory depression, apnoea, muscular rigidity and bradycardia. If these remain untreated respiratory arrest, circulatory depression, or cardiac arrest could occur. Respiratory depression is more likely to occur with intravenous administration if a dose is given too rapidly; it rarely occurs with intramuscular administration. If respiratory depression occurs during anaesthesia, assisted or controlled respiration will provide adequate ventilation without reversing analgesia. Respiratory depression can be immediately reversed by narcotic antagonists (e.g. nalorphine) which, it should be noted, will also reverse analgesia.

Muscular rigidity may be associated with reduced pulmonary compliance and/or apnoea, laryngospasm or bronchospasm.

Prompt reversal of this side effect can be achieved with the intravenous administration of an appropriate single dose of a muscle relaxant such as suxamethonium. Assisted or controlled respiration is required to provide ventilation after the use of muscle relaxants.

Bradycardia and other cholinergic effects may occur and can be controlled with the appropriate dose of atropine. The inclusion of atropine or other anticholinergic agents in the pre-anaesthetic regimen tends to reduce the occurrence of such effects.

Less common: Hypotension, hypertension, dizziness, blurred vision, miosis, nausea, emesis, laryngospasm, diaphoresis, itching and euphoria. Motor stimulation and bronchospasm may occur with high doses of fentanyl.

Fentanyl/neuroleptic combination: When a neuroleptic such as droperidol is used with fentanyl, the following adverse reactions can occur: chills and/or shivering, restlessness and postoperative hallucinatory episodes sometimes associated with transient period of mental depression: extrapyramidal symptoms (dystonia, akathisia and oculogyric crisis) have been observed up to 24 hours postoperatively.

When they occur, extrapyramidal symptoms can usually be controlled with antiparkinson agents. Postoperative drowsiness is also frequently reported following the use of droperidol.

Elevated blood pressure with or without pre-existing hypertension, has been reported following administration of fentanyl combined with droperidol. This might be due to unexplained alterations in sympathetic activity following large doses; however, it is also frequently attributed to anaesthetic and surgical stimulation during light anaesthesia.

Postmarketing Experience:

Serotonin syndrome (See Warning and Precautions)

Adrenal insufficiency (See Warnings and Precautions)

Androgen deficiency: Cases of androgen deficiency have occurred with chronic use of opioids. Chronic use of opioids may influence the hypothalamic-pituitary-gonadal axis, leading to androgen deficiency that may manifest as low libido, impotence, erectile dysfunction, amenorrhea, or infertility. The causal role of opioids in the clinical syndrome of hypogonadism is unknown because the various medical, physical, lifestyle, and psychological stressors that may influence gonadal hormone levels have not been adequately controlled for in studies conducted to date. Patients presenting with symptoms of androgen deficiency should undergo laboratory evaluation.

Infertility: Chronic use of opioids may cause reduced fertility in females and males of reproductive potential. It is not known whether these effects on fertility are reversible.

SYMPTOMS AND TREATMENT OF OVERDOSE:

Symptoms: narcosis (which may be preceded by marked skeletal rigidity, cardiorespiratory depression accompanied by cyanosis, followed by a fall in body temperature, circulatory collapse, coma and possibly death.

Treatment: In the presence of hypoventilation or apnoea, oxygen should be administered and respiration assisted or controlled as necessary. A patent airway must be maintained.

If depressed respiration is associated with muscular rigidity, an intravenous intramuscular blocking agent might be required to facilitate assisted or controlled respiration.

The patient should be carefully observed for 24 hours; body warmth and adequate fluid intake should be maintained. If severe or persistent hypotension occurs, the possibility of hypovolaemia should be considered and managed with appropriate parenteral fluid therapy.

A specific narcotic antagonist, such as nalorphine or naloxone, should be available for use as indicated to manage respiratory depression. This does not preclude the use of more immediate countermeasures. The duration of respiratory depression following overdosage of fentanyl is usually longer than the duration of narcotic antagonist action.

INCOMPATIBILITIES:

Fentanyl continuous or intermittent in Glucose 5% or sodium chloride 0.9%. Fentanyl not be combined with other aqueous solutions of medicaments except with water for injection. The injection should be freshly prepared and unused portions should be discarded.

STORAGE CONDITIONS:

Store below 30ºC. Protect from light.
KEEP OUT OF REACH OF CHILDREN. JAUHKAN DARIPADA KANAK-KANAK.

SHELF LIFE:
Please refer to outer package.

PACK SIZE:
Pack in 2 ml ampoules in box of 10's only.

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD
Lot 2599, Jalan Seruling 59 Kawasan 3,
Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA.